

included 24 modules to route fault-insertion signals (one module with 30-amp switching, 22 modules at 20 amperes and one module at 2 amperes) plus programmable resistor cards for sensor simulation. Other cards for FPGA functionality and CAN/LIN communication protocol support were also included. In addition to the cost and flexibility advantages already mentioned, the application of PXI solutions also improved the test system's performance and cabling elegance in comparison to the original proprietary system.

2 Satellite payload testing

Customer. A satellite company delivering operational space missions for a range of applications including earth observation, science and communications. The company designs, manufactures and operates high-performance satellites and ground systems for a fraction of the price normally associated with space missions. These cost savings are achieved by implementing commercial off-the-shelf (COTS) satellite technology. This process takes standard consumer technologies such as those used in personal computers, and adapts these to the unique environment of space.

Application. Satellite payload testing is critical for several reasons. First, as the hardware is extremely expensive, a lot of testing is expected. Second, if a failure occurs after launch, it is extremely difficult, but not impossible, to return the entire product for service.

Satellites and payloads are complex systems made up from multiple mixed-technology sub-systems. The traditional approach to satellite development is a very linear fashion—specify everything upfront, design to specifications, test the prototype, then assemble and verify the flight unit. This approach is time-consuming and unsuitable for rapid, low-cost developments, which require more agile methodologies.

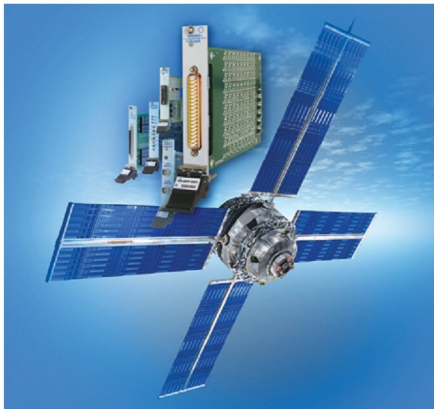


Fig. 3: PXI can simulate the numerous resistive temperature devices located in a satellite system and show how the satellite control systems react to an alarm

The use of simulation can provide major improvements on cost and time-scale. Using COTS components and standard tool-sets, a sub-system simulator can be developed and tested far in advance of flight-grade equipment, often at a much lower cost. This yields two major advantages to the project:

1. The designs are developed and progressed far in advance of the main development. This allows early detection and resolution of inherent design issues avoiding any costly rework of flight equipment.

2. Interface simulators can be used for system integration-testing, far in advance to identify and resolve sub-system integration issues. This is especially important for large-scale programmes where a single satellite may be made by many different sub-contracted manufacturers. Unlike hardware, the simulator is open, providing a much higher degree of debugging ability.

Solution. When specifying hardware for this latest simulation requirement, the customer chose not to base it on heritage tools and strategies, instead took a fresh look at what was available and possible. The goal was to find an optimally-engineered solution within a defined budget.

It was decided to base the hardware primarily upon the PXI standard. The reasoning behind this came down to one statement from the customer: "The solution fitted our requirement, as opposed to our requirement having to fit the solution."

In the case of thermal control system, which monitors key temperatures within the satellite, PXI high-density precision resistor module is a key test product. Six modules were used to simulate 36 space-grade thermistors across their full range of -80°C to $+55^{\circ}\text{C}$. These PXI precision resistor modules could cover the entire range of the thermistors requiring simulation.